



Building RAG Agents with LLMs

Recent RAG Research:

From Searching Once to Searching with a Plan

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본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

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Five Papers, Three Research Questions

Learning to search, learning from steps, working on the web

ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning

Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning

Process vs. Outcome Reward: Which is Better for Agentic RAG Reinforcement Learning

WebThinker: Empowering Large Reasoning Models with Deep Research Capability

DeepResearcher: Scaling Deep Research via Reinforcement Learning in Real-world Environments

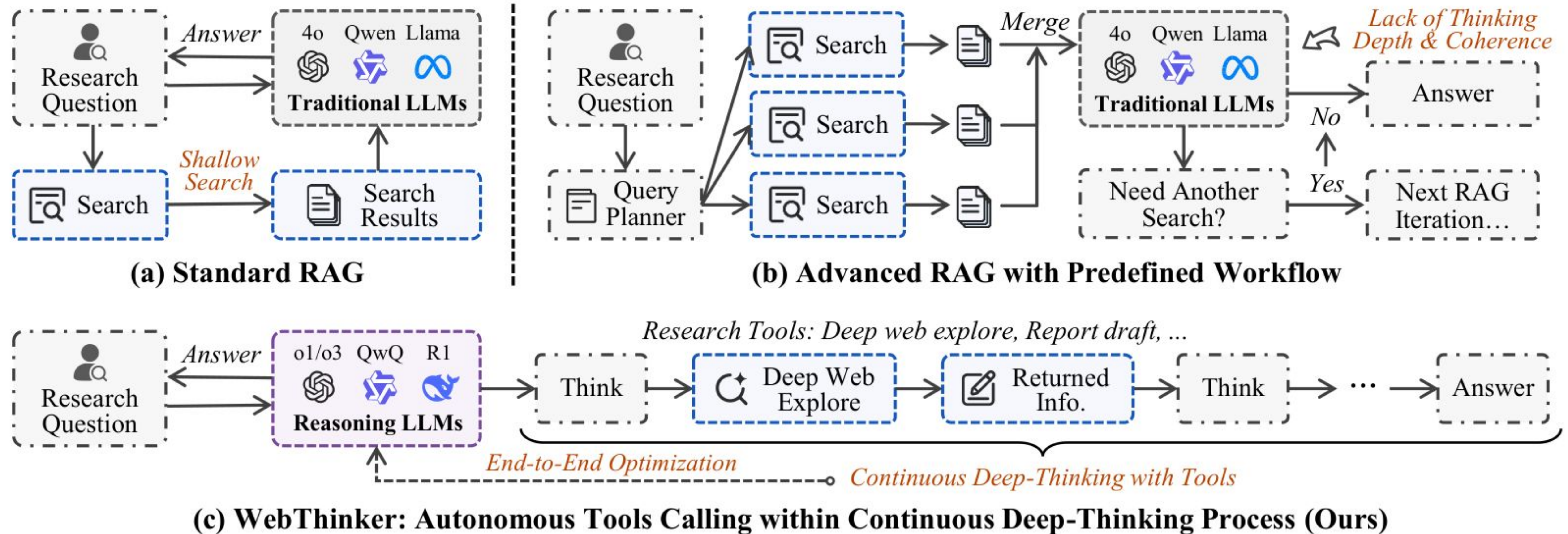
ReSearch / Search-R1: how can search become a learned action?

ReasonRAG: which intermediate choices deserve feedback?

WebThinker / DeepResearcher: what changes on the real web?

RAG and Autonomous Search

RAG uses retrieved evidence to support an answer



Top: a preset retrieval workflow

Bottom: the model chooses tools while reasoning

A Question with Two Missing Facts

An actual question from the ReSearch paper

Input Question



Who was president of the United States in the year that Citibank was founded?

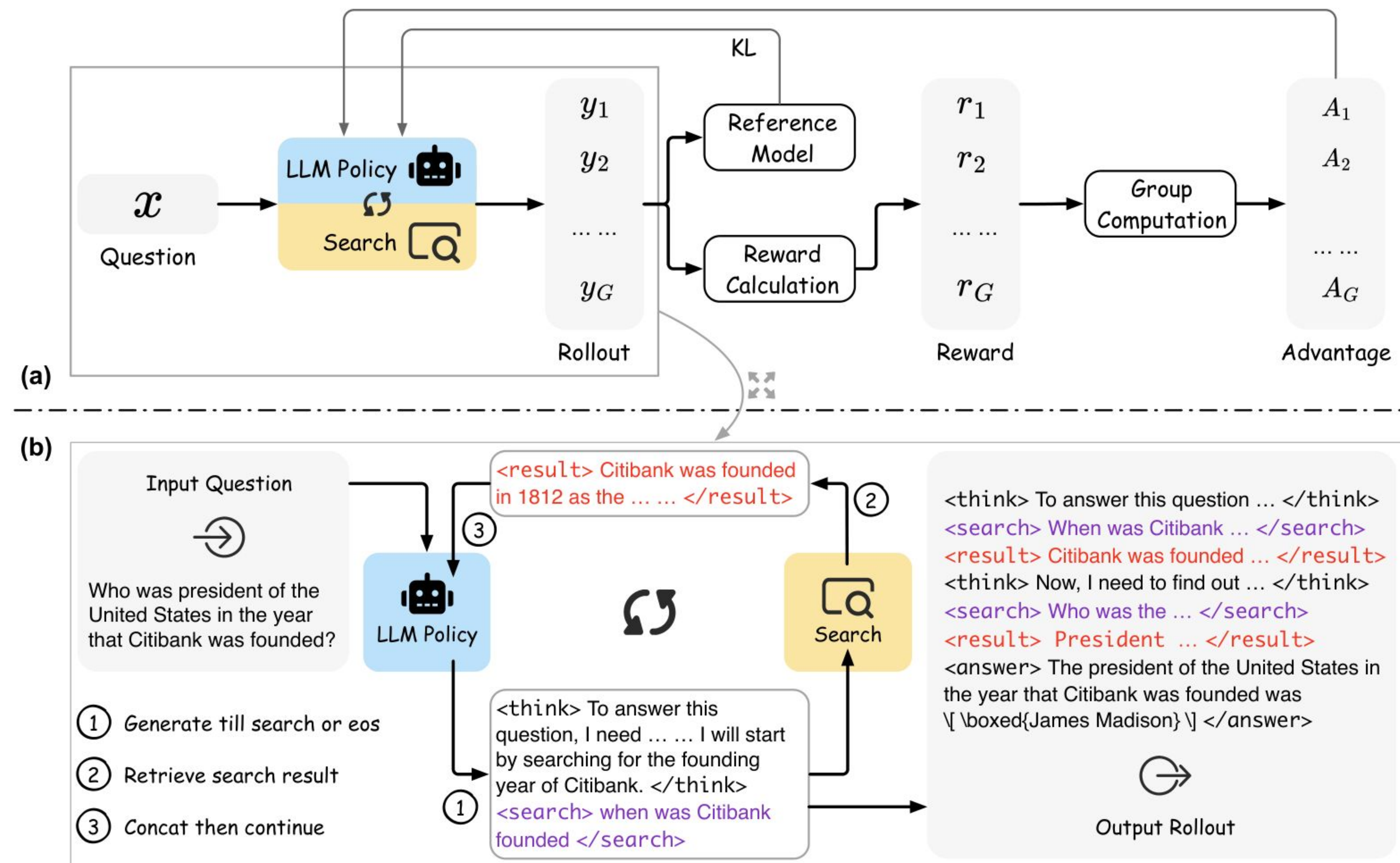
First fact: the founding year of Citibank

Second fact: the US president in that year

A result from one search becomes the next query.

ReSearch: Learning with Search

The upper panel trains the model; the lower panel generates one attempt

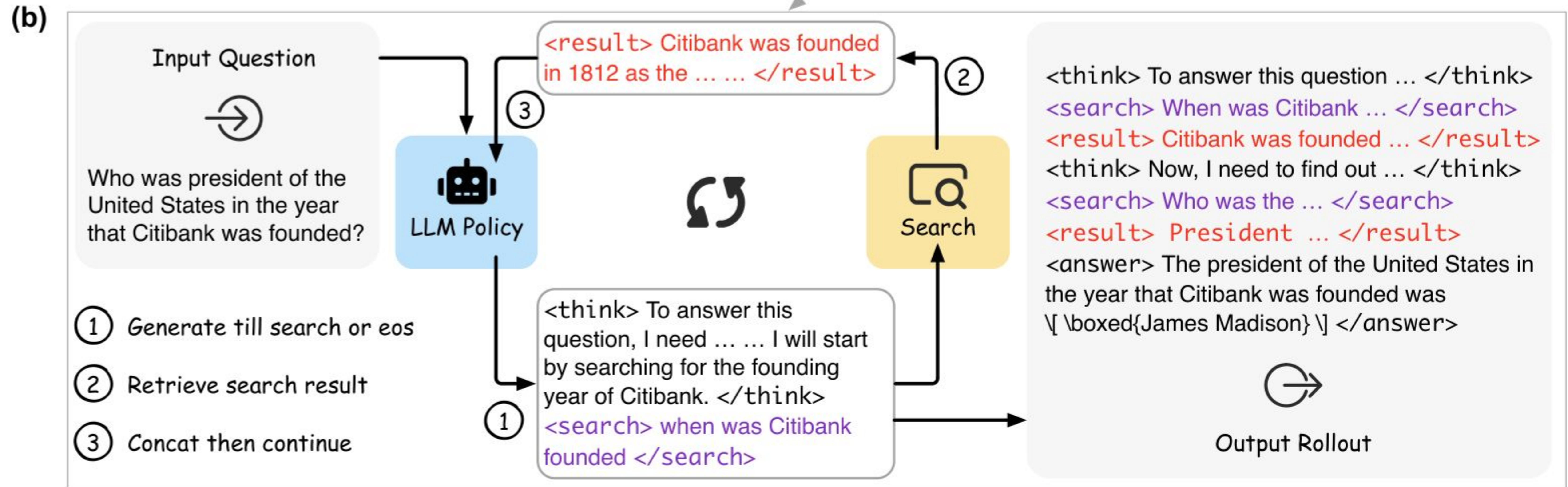


Rollout: one complete attempt at answering

Reward: a score used to improve the policy

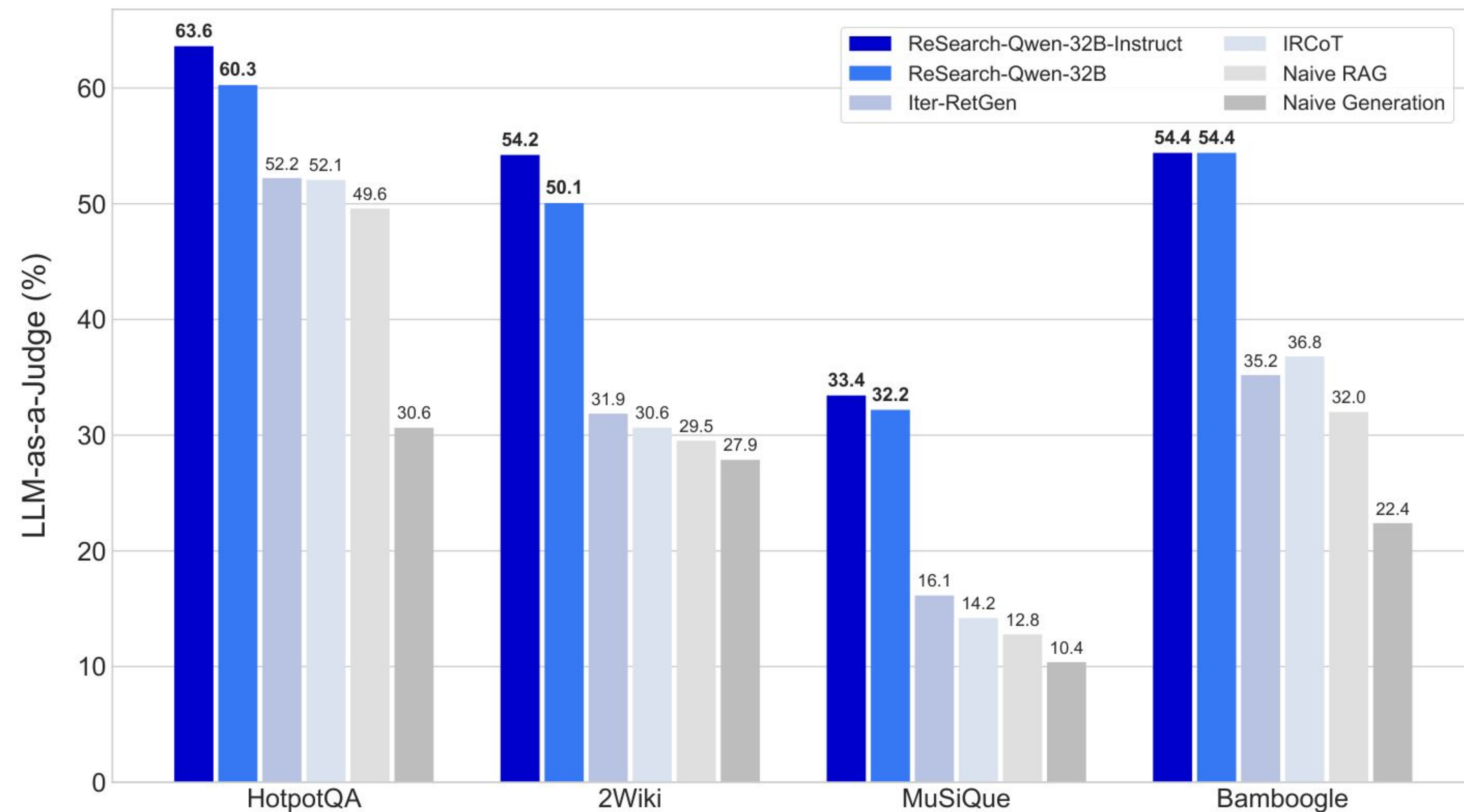
ReSearch: A Search Result Changes the Next Step

The paper's rollout, enlarged



ReSearch: Results across Multi-hop Benchmarks

The original comparison chart

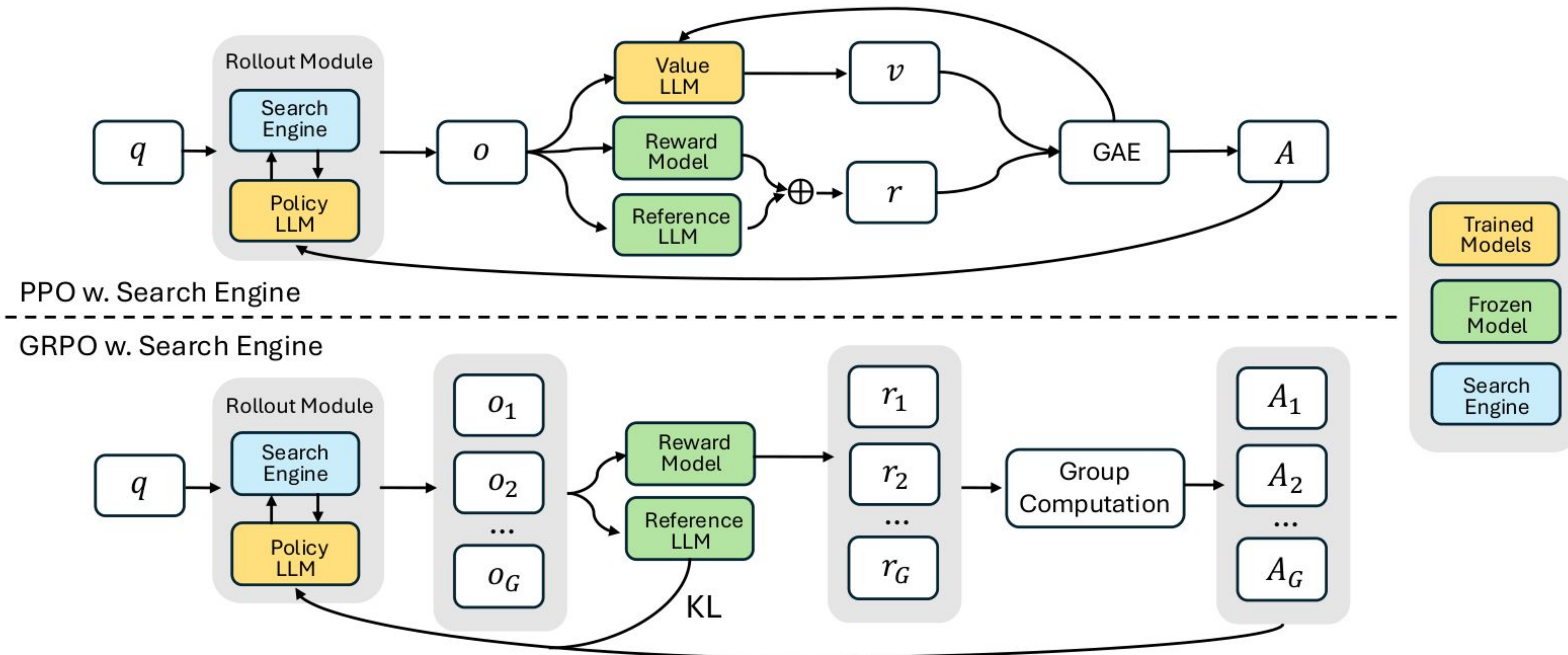


Read the task names before comparing bars

The vertical axis is LLM-as-a-Judge, not Exact Match

Search-R1: Search inside RL Training

PPO and GRPO both generate answers with a search engine



Blue: search engine inside the rollout

Yellow: policy model to train

Search-R1: Reading the Results Table

Table 2, Qwen2.5-7B block

Methods	General QA				Multi-Hop QA			
	NQ [†]	TriviaQA [*]	PopQA [*]	HotpotQA [†]	2wiki [*]	Musique [*]	Bamboogle [*]	Avg.
Qwen2.5-7b-Base/Instruct								
Direct Inference	0.134	0.408	0.140	0.183	0.250	0.031	0.120	0.181
CoT	0.048	0.185	0.054	0.092	0.111	0.022	0.232	0.106
IRCoT	0.224	0.478	0.301	0.133	0.149	0.072	0.224	0.239
Search-o1	0.151	0.443	0.131	0.187	0.176	0.058	0.296	0.206
RAG	0.349	0.585	0.392	0.299	0.235	0.058	0.208	0.304
SFT	0.318	0.354	0.121	0.217	0.259	0.066	0.112	0.207
R1-base	0.297	0.539	0.202	0.242	0.273	0.083	0.296	0.276
R1-instruct	0.270	0.537	0.199	0.237	0.292	0.072	0.293	0.271
Rejection Sampling	0.360	0.592	0.380	0.331	0.296	0.123	0.355	0.348
Search-R1-base	0.480	0.638	0.457	0.433	0.382	0.196	0.432	0.431
Search-R1-instruct	0.393	0.610	0.397	0.370	0.414	0.146	0.368	0.385

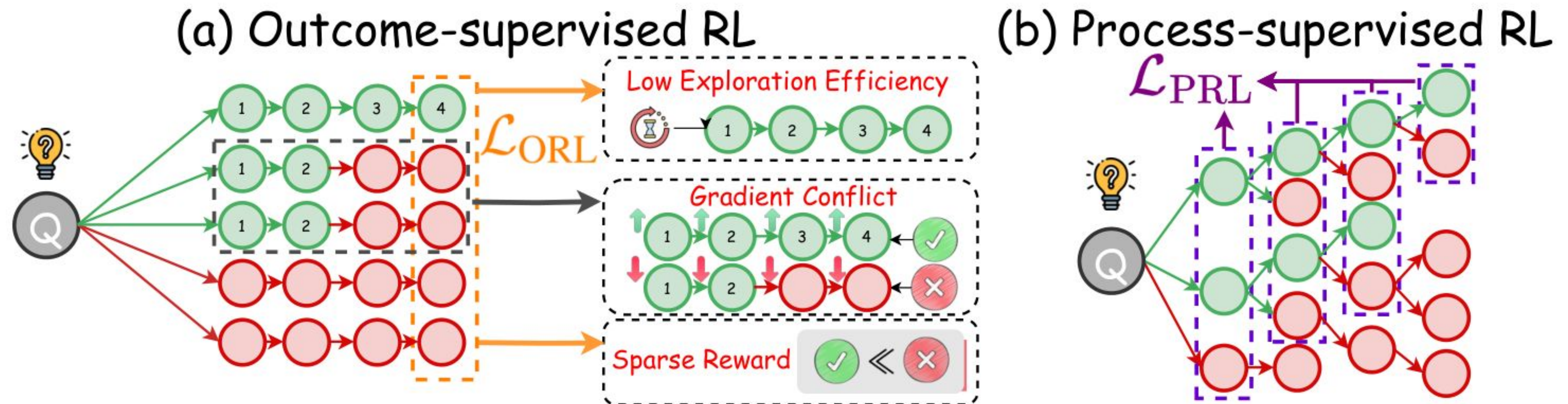
RAG: 0.304 average EM

Search-R1-base: 0.431 average EM

Same table, different training setups. This is not a controlled single-change ablation.

Outcome Feedback and Process Feedback

ReasonRAG compares feedback at different levels

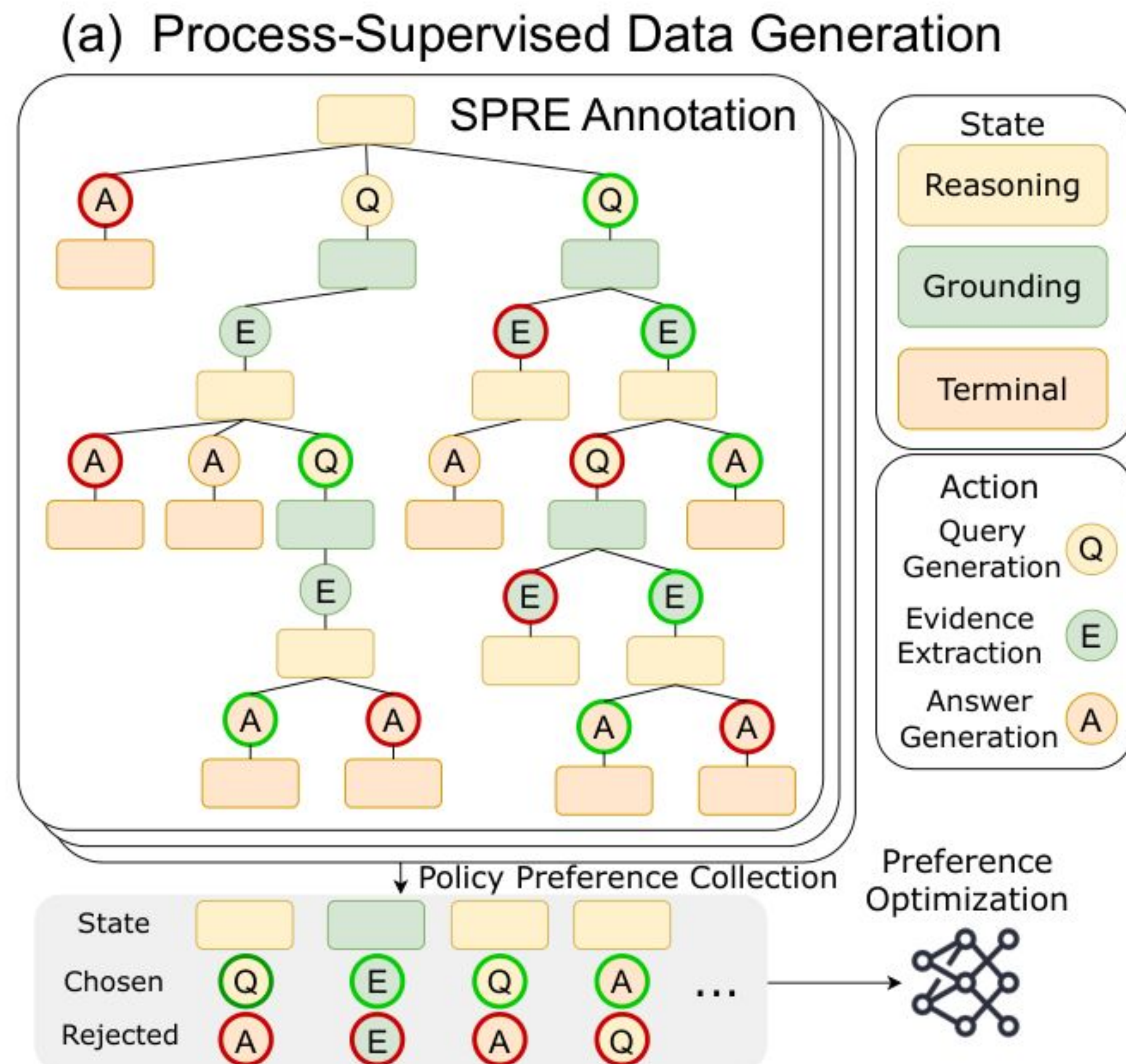


Outcome: assess the final result

Process: assess intermediate choices

ReasonRAG: Exploring Alternative Actions

A tree of candidate actions and estimated process rewards



Branch: another possible next action

Color: an estimated process reward

Preference pairs: chosen and rejected actions

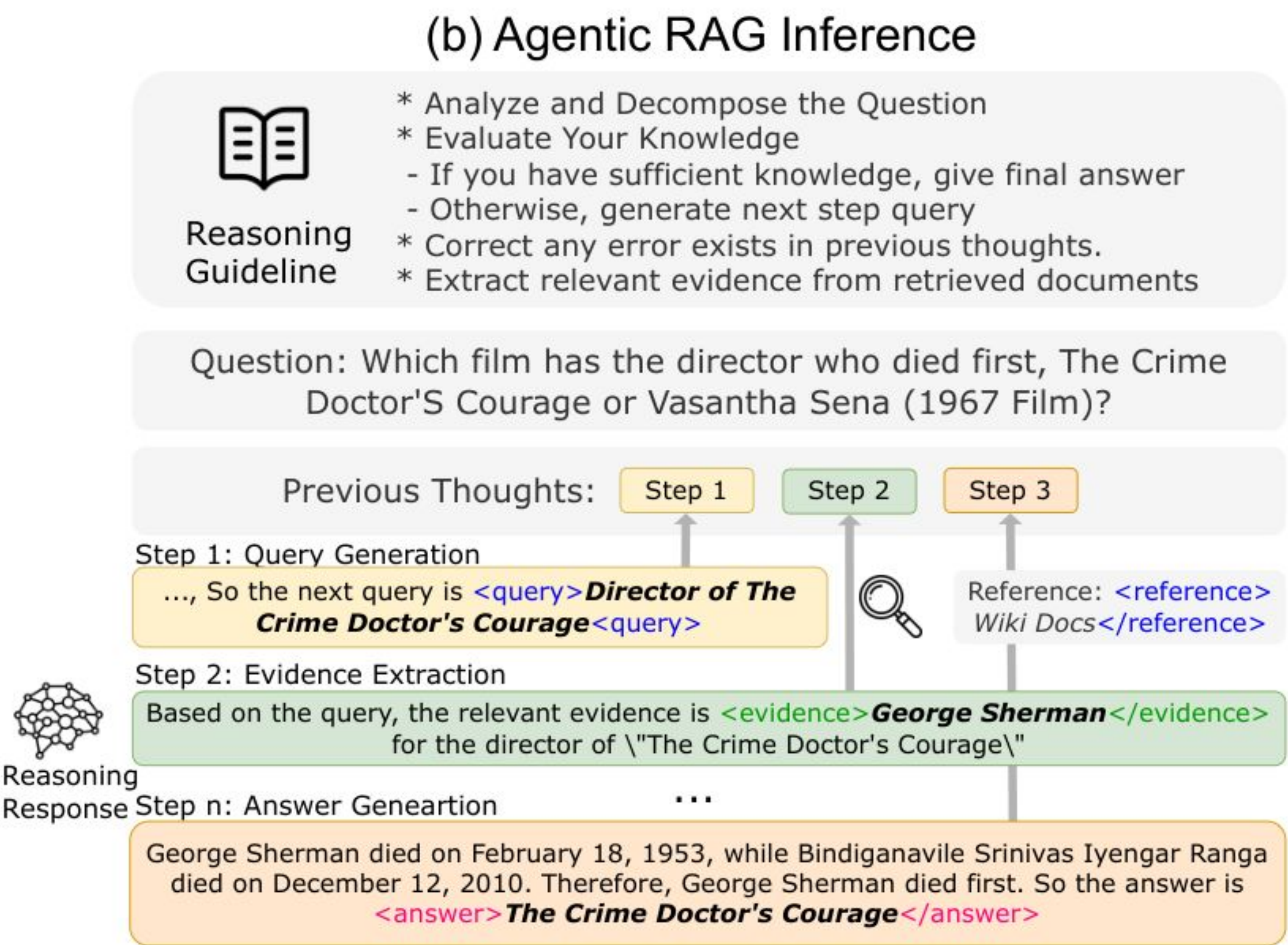
ReasonRAG: Query, Evidence, Answer

The paper's inference example

Generate a query

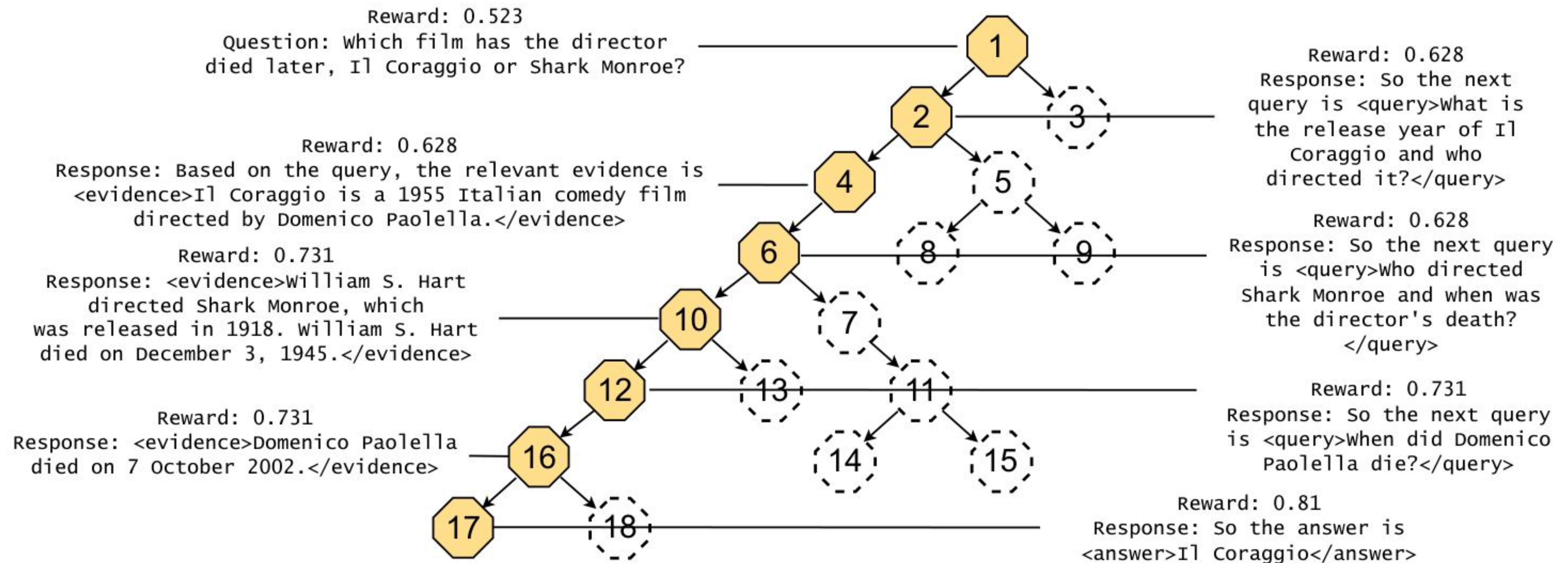
Extract relevant evidence

Answer when the evidence is enough



ReasonRAG: A Concrete Search Tree

An example of process-level preference annotation



Follow the selected path through the numbered nodes

A useful action should lead toward a correct answer

ReasonRAG: Results and Training Data

Table 2 reports both EM and F1

Type	Method	PopQA		HotpotQA		2WikiMulti		Bamboogle		MuSiQue		Avg.	
		EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁
Zero-shot	Naïve Generation	12.7	16.5	15.7	24.8	20.2	28.0	6.4	17.4	2.7	10.2	11.5	19.4
	Standard RAG	38.4	44.7	29.3	39.9	29.4	36.3	17.6	24.1	6.7	15.1	24.3	32.0
Active	FLARE	14.3	17.6	18.1	25.7	27.9	32.8	12.0	20.8	4.3	12.6	15.3	21.9
	Self-RAG(146k)	22.7	33.9	21.0	29.7	12.0	25.2	1.6	10.9	4.6	13.3	12.4	22.6
Adaptive	AdaptiveRAG(3k)	36.6	41.5	29.1	40.7	24.2	33.4	18.4	26.1	6.9	14.3	23.0	31.2
RAG-CoT	Iter-Retgen	38.7	44.9	30.3	42.1	31.2	38.7	19.2	26.4	7.7	14.2	25.4	33.3
	IRCoT	36.2	43.6	27.7	41.5	23.5	32.5	17.2	22.5	8.6	13.2	22.6	30.7
Summary	RECOMP	<u>40.5</u>	<u>45.8</u>	29.7	41.2	33.2	39.4	21.7	28.6	9.2	15.8	26.9	34.2
	LongLLMLingua	39.2	45.1	31.4	43.2	34.5	40.2	20.3	27.4	8.7	14.9	26.8	34.2
	Selective-Context	34.9	41.5	19.3	27.3	20.3	29.7	15.3	22.6	6.1	13.7	19.2	27.0
Reasoning	Search-o1	33.2	40.3	24.8	38.1	16.4	27.1	30.4	40.6	6.3	13.7	22.2	31.96
	AutoRAG(10k)	38.6	44.1	33.3	43.7	39.5	46.1	24.8	32.2	11.3	18.3	29.5	36.9
	Search-R1(90k)	39.7	44.8	<u>37.0</u>	<u>47.0</u>	<u>41.4</u>	<u>48.0</u>	<u>32.0</u>	<u>43.8</u>	14.6	<u>19.9</u>	<u>32.8</u>	<u>40.7</u>
	ReasonRAG(5k)	41.5*	46.2*	38.4*	48.9*	43.6*	50.4*	36.0*	45.5*	<u>12.8</u>	20.6*	34.4*	42.3*

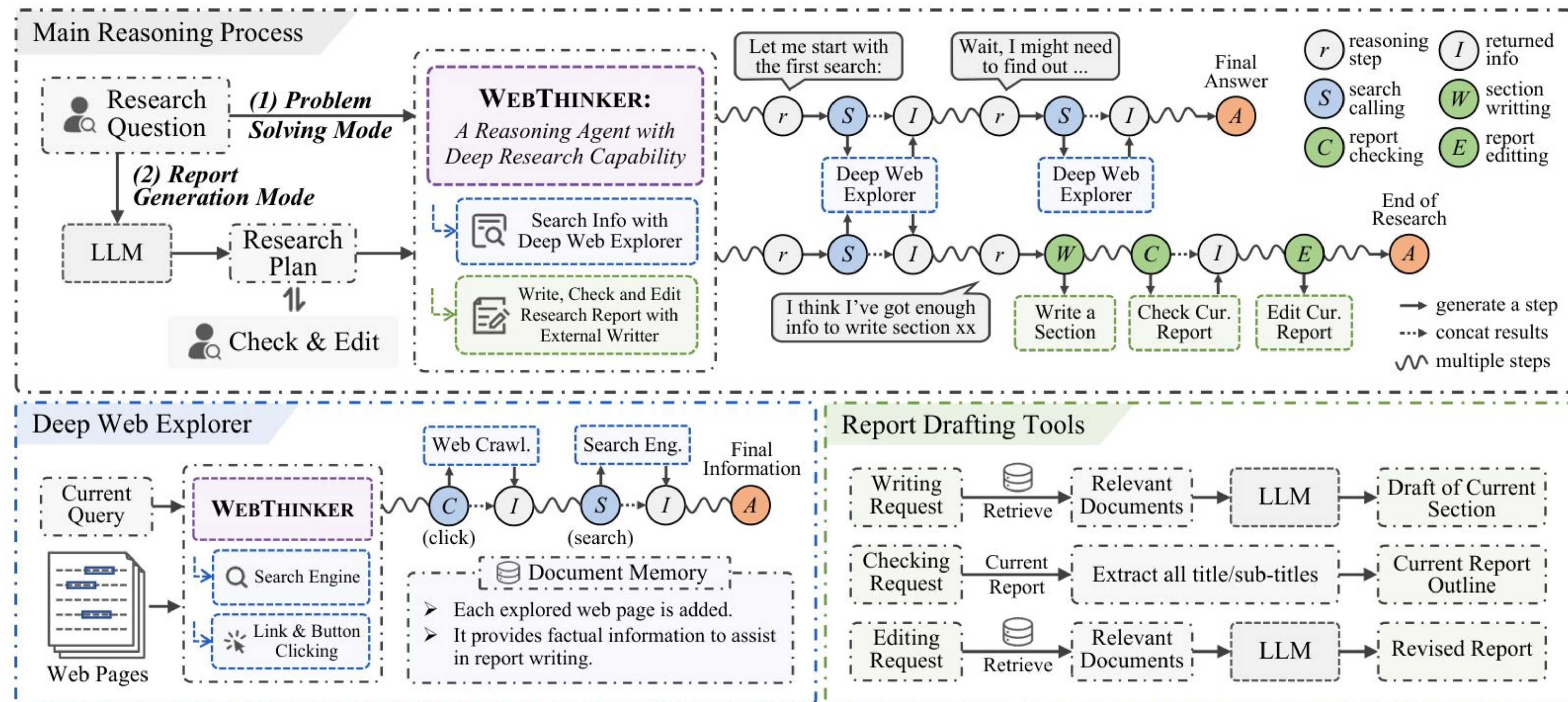
Search-R1 (90k): 32.8 EM, 40.7 F1

ReasonRAG (5k): 34.4 EM, 42.3 F1

Training-question count is not total training cost.

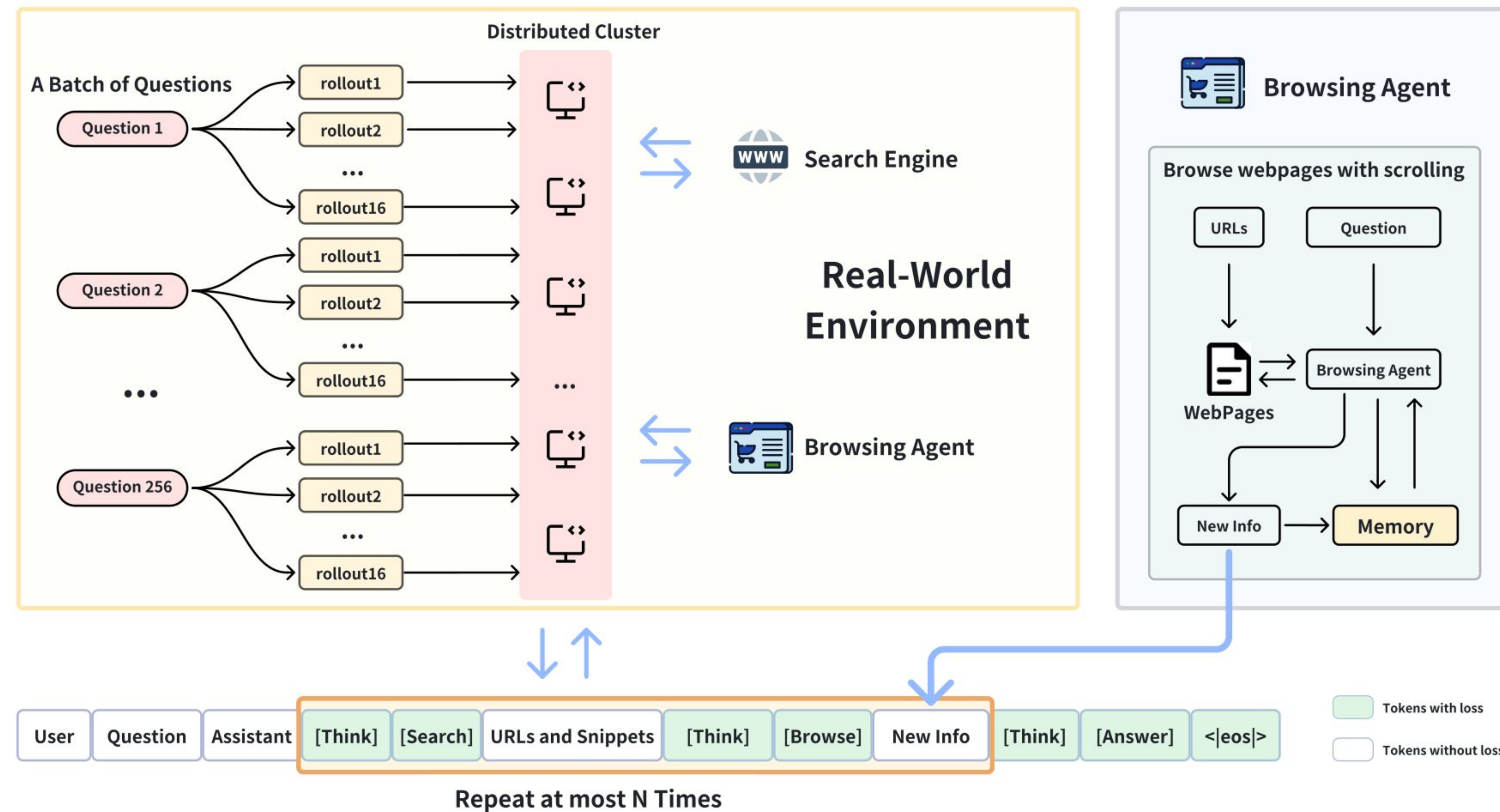
WebThinker: A Research Workflow

Problem solving and report generation use different tool paths



DeepResearcher: Learning on the Real Web

Search and browsing interact with a real environment



DeepResearcher: Planning before Search

A real case-study excerpt

Planning

Question: *What is the name of the famous bridge in the birthplace of Gloria in D Major's composer?*

Ground Truth: Rialto Bridge

Response Trajectory from DeepResearcher:

`<think>`

To answer this question, I need to figure out who Gloria in D Major's composer was and their birthplace. Then I can find the famous bridge in that city.

Step 1: Identify the composer of Gloria in D Major.

Step 2: Find the place of birth of that composer.

Step 3: Look for a famous bridge in that city.

`</think>`

`<search>`

composer of Gloria in D Major

birthplace of the composer of Gloria in D Major

`</search>`

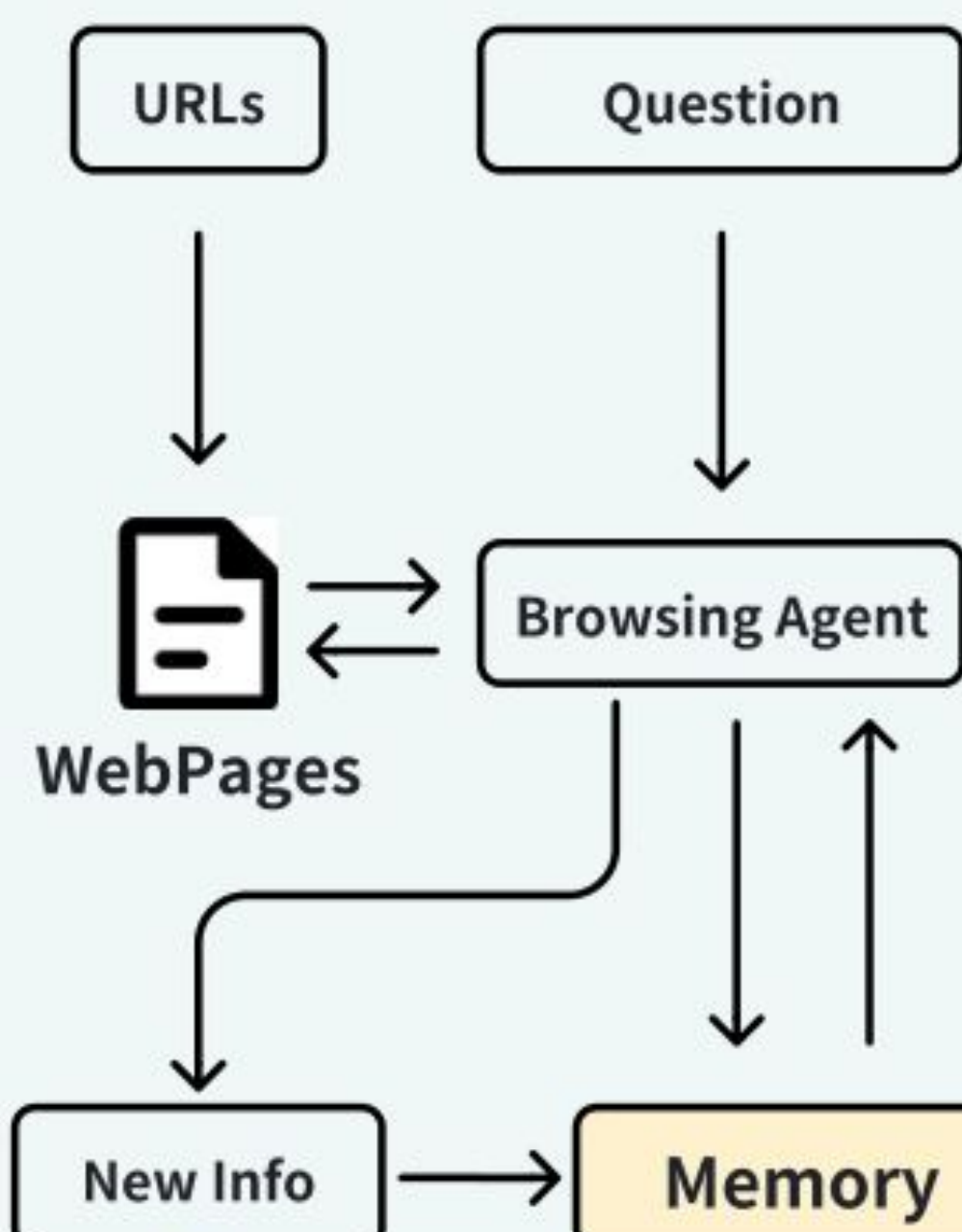
Make a plan first

Plan: composer, birthplace, then a bridge



Browsing Agent

Browse webpages with scrolling



Browse: open pages and update memory

DeepResearcher: Reflection and Uncertainty

The paper shows correction and an explicit lack of evidence

<think>

The web search results provide clues about Djedefhor's father. **However, the search results are mostly about Boromir's father Denethor from the Lord of the Rings, not Djedefhor's father. I need to refine my search to find information about Djedefhor specifically.**

</think>

<search>

who was Djedefhor's father and where was he buried

Djedefhor father name

Djedefhor father identity

</search>

<tool_response>

...(omitted to save space)

</tool_response>

Reflection and adjust
research direction

It seems the information about the oil production of Luanda is not directly available in the search results or in the previously browsed webpages. Given the information from previous searches, Angola is a major oil producer, and as the capital and largest city, Luanda likely plays a key role in the country's oil production. However, to answer the question precisely, we may need a more detailed report on the city's oil production, which might be included in specialized databases or recent reports. **Since the exact oil production of Luanda is not available from the searches and browsing,** and given the context and the previously found information that Angola is a major oil producer, **it's reasonable to infer that the oil production in the city at the mouth of the Congo River (Luanda) is a significant part** of Angola's total oil production. **However, without a precise figure, I will not provide a number.**

</think>

<answer>

significant portion

</answer>

Be honest when didn't
find the exact answer

Reflection: revise the search direction

Honesty: avoid inventing a precise number

DeepResearcher: Results beyond Training Domains

Table 2 compares three out-of-domain benchmarks

Method	Inference Environment	Musique		Bamboogle		PopQA	
		F1	MBE	F1	MBE	F1	MBE
<i>Prompt Based</i>							
CoT	Local RAG	8.5	7.4	22.1	21.6	17.0	15.0
CoT + RAG	Local RAG	10.0	10.0	25.4	27.2	46.9	48.8
Search-o1*	Local RAG	16.8	21.3	35.8	38.4	36.9	42.4
Search-o1	Web Search	14.7	19.7	46.6	53.6	38.3	43.4
ReAct-style Agent	Web Search	8.9	10.0	34.4	36.8	19.1	20.5
<i>Training Based</i>							
Search-r1-base	Local RAG	26.7	27.5	56.5	57.6	43.2	47.0
Search-r1-instruct	Local RAG	26.5	28.3	45.0	47.2	43.0	44.5
R1-Searcher	Web Search	22.8	25.6	64.8	65.6	42.7	43.4
DeepResearcher (Local RAG)	Local RAG	12.7	12.5	42.7	46.4	23.2	23.4
DeepResearcher	Web Search	27.1	29.3	71.0	72.8	48.5	52.7

Read the inference environment column

Compare F1 and MBE separately

Out-of-domain gains do not prove that every web task improves.



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